

ASSIGNMENT-AR/VR



I used 3js elements for mainly building the 3d environment, camera and rendering some textures

- **THREE.Scene**: Used for environment building like walls, floor and flashcards.
- **THREE.PerspectiveCamera**: Used for Positioning like human eye level
- **THREE.WebGLRenderer**: For rendering the glass textures

For lighting, I used,

- **AmbientLight** which provides low-intensity light
- **SpotLights** which is for flashcards to give a spotlight.
- Also used a LED strip which is basically cylinder with **PointLights**.

Also used some external 3d assets which are chandelier and potted plant from **Sketchfab** using **GLTFLoader**

This project provides fully immersive first-person experience

- Using **PointerLockControls**, the user can look around 360 degree and move using WASD keys.

Flashcards are placed in a glass wall using fixed coordinates. When clicked, a invisible laser like thing called **Raycaster** identifies the card, and **GSAP** triggers a flip animation on a randomly chosen X or Y axis using `math.random()`. GSAP library is a used for smooth animations.

The cards don't end up upside down because the code detects the flip direction. If it's a top-to-bottom flip, it applies a 180 degree correction to the text's local Z-axis so it lands right-side up. This is what I understand